bio

12 · CBSE

Based on 5 year(s) of previous papers

Topic Priority Overview

Topic	Priority	Confidence	Freq	Marks
Microspore formation: meiosis of pollen mother cell	High	High	3×	MCQs across papers (1-mark each); appears repeatedly in
Stop codon identification (UGA as stop codon)	High	High	4×	MCQs (1 mark each) across multiple papers
Chargaff's rule / base proportion in double-stranded DNA	High	High	4×	MCQs and short computations (1-2 marks)
IUD types: Multiload 375 and copper IUDs (named brand mention)	Medium	Medium	2×	MCQ mentions (1 mark) and short-answer mentions (2 marks in some papers)
RNA as catalytic molecule (ribozymes) / first genetic material evidence	Medium	Medium	2×	MCQ recognition (1 mark) and short conceptual mention
lac operon regulation by repressor (negative regulation)	High	High	4×	MCQ & short-answer contexts (1-3 marks); repeated concept
Plasmid: definition as extrachromosomal circular DNA	High	High	4×	MCQs and cloning/ biotech long Qs (1-5 marks)
Restriction enzymes: recognition site naming and sticky/blunt ends	High	High	4×	MCQs, short questions and diagram-based (1-3 marks)
Replication fork: definition and role in replication of long DNA	High	High	4×	Short-answer and conceptual (3 marks frequently)
DNA polymorphism: definition and two applications	High	High	3×	Short-answer (3 marks) repeated
PCR / molecular detection at low pathogen concentration	High	High	4×	Short-answer 3 marks and technique questions (Section C/E)
Baculoviruses as biological control: three reasons they are preferred	Medium	Medium	2×	Short-answer 3-mark question when asked
Hardy-Weinberg equilibrium: definition and factors (Founder effect, gene migration)	High	High	4×	Case-based question in Section D (4 marks) often with subparts
Population logistic growth model: equation and K carrying capacity	High	High	4×	Short/long questions including numeric calculations (2-5 marks)
Decomposition to humus: stepwise process (fragmentation, leaching, catabolism, mineralisation, humification)	Medium	Medium	2×	Short-answer 2-3 marks

Uterus wall structure: layers and functions of any two layers	High	High	3×	Short 3-mark question often asking layering + functions
Trophoblast function in human embryo	Medium	Medium	2×	Very short answer (2 marks) with function
Ploidy of primary and secondary spermatocytes	High	High	3×	Very short answer (2 marks) across papers
Polygenic inheritance deviation from Mendelian patterns	Medium	Medium	2×	Very short answer (2 marks)
Blood groups: non-Mendelian patterns (multiple allelism and co-dominance)	High	High	4×	Short answers and concept questions (2-3 marks) frequently
Tears as immune barrier: chemical barrier example and analogues	Medium	Medium	2×	Very short answer 2 marks
Methods to introduce alien DNA into animal and plant cells (electroporation, biolistic/gene gun)	High	High	4×	Short-answer 2-3 marks and Section E technique questions
Adaptive radiation / divergent evolution identification and examples	Medium	Medium	2×	MCQ identification and short explanation (1-3 marks)
Pedigree chart interpretation: inheritance pattern identification (X-linked, autosomal)	High	High	4×	MCQ pedigree identification and longer genetics questions (1-3 marks)
Vaccination long-term immunity: immunological memory and anamnestic response	High	High	4×	Case-based 4-mark question in immunity section plus short
Angiosperm seed advantages and polyembryony (define and example)	High	High	3×	Section E long question (5 marks) often multipart
Cloning vector pBR322: diagram labelling and role of 'rop' gene	Medium	Medium	2×	Section E 5-mark long question (labelling + roles + consequences)
AIDS: cells attacked (CD4+ T cells), diagnostic tests (ELISA), and WHO prevention steps	High	High	3×	Case-based 4-mark question (short subparts) and MCQ references
Spermatogenesis: stages, hormones (FSH, LH) and cell-stage mapping	High	High	4×	Section E long questions and short-answer parts (2-5
Predator vs prey: distinction and ecological roles of predators	Medium	Medium	3×	Short-answer 3 marks asking distinction + roles
In situ vs ex situ conservation: differences and examples	Medium	Medium	2×	Short 5-mark or 2-5 mark comparisons in Section E/C
Events in mosquito after sucking infected blood	High	High	1×	Short-answer 2 marks (often part of Section D case question)
Species-Area relationship and logistic growth equation	High	High	2×	Long/short answer (2-5 marks); graphical questions

Detailed Study Notes

Read these notes carefully — they're based on actual patterns found in your past papers.

1. Microspore formation: meiosis of pollen mother cell

High Conf. High Priority

Appeared in: Paper 1 · Paper 3 · Paper 5 · 3× total

Question types: MCQ: 3

Kya Padhna Hai:

- 'Microspores are formed by : Meiosis of pollen mother cell' (identify process and cell type)
- difference between microspore formation and megaspore formation when asked in MCQ wording
- events of meiosis specifically in pollen mother cell leading to microspores
- usage of term 'microspore' and its ploidy (n) as implied in MCQ stem

Kaise Poochha Jaata Hai:

Mostly 1-mark MCQ phrasing like 'Microspores are formed by : (A) Meiosis of pollen mother cell ...' Occasionally mentioned within longer embryo sac/angiosperm development questions.

Repeat Pattern:

Same MCQ ('Microspores are formed by') appears in Paper 1 and Paper 3 and equivalent concept in Paper 5 — consistently tested as a crisp MCQ item.

Key Terms:

- *microspore*
- *pollen mother cell*
- *meiosis*
- *haploid*

2. Stop codon identification (UGA as stop codon)

High Conf. High Priority

Appeared in: Paper 1 · Paper 3 · Paper 2 · Paper 5 · 4× total

Question types: MCQ: 4

Kya Padhna Hai:

- Recognise stop codons with example UGA as asked: 'Which of the following is a Stop codon ? (UGA)'
- Compare UUU (Phe), UGA (Stop), UGU (Cys), UAC (Tyr) as per MCQ option sets
- Recall basic genetic code mapping frequently used in MCQs
- Know that stop codons terminate translation (brief phrasing in MCQ context)

Kaise Poochha Jaata Hai:

Direct MCQ: one-line stem asking 'Which of the following options is a Stop codon ?' with four choices.

Repeat Pattern:

Stop-codon identification recurs across at least 3-4 papers in identical MCQ format !' high predictability.

Key Terms:

- UGA
- stop codon
- genetic code
- translation termination

3. Chargaff's rule / base proportion in double-stranded DNA

High Conf. High Priority

Appeared in: Paper 1 · Paper 2 · Paper 3 · Paper 4 · 4× total

Question types: MCQ: 3 | Short: 2

Kya Padhna Hai:

- Chargaff's observation: 'proportion of bases in a double-stranded DNA' and scientist name Erwin Chargaff as answer in MCQ
- Percentage relations: A=T and G=C; compute %C when %A given (example: 'If adenine constitutes 31% ... percentage of cytosine')
- Use simple arithmetic: %A + %T + %G + %C =100 and A=T
- Typical MCQ/short-answer numeric framing seen in papers

Kaise Poochha Jaata Hai:

Either one-line MCQ asking 'proportion of bases in a double-stranded DNA ?' (scientist) or short calculation: 'If adenine is 31%... what is cytosine %?'

Repeat Pattern:

Chargaff/base composition numeric problems and related MCQs appear repeatedly across papers !' highly reliable to revise.

Key Terms:

- Chargaff
- A=T
- % base composition
- base pairing

4. lac operon regulation by repressor (negative regulation)

High Conf. High Priority

Appeared in: Paper 1 · Paper 2 · Paper 3 · Paper 4 · 4× total

Question types: MCQ: 3 | Short: 2

Kya Padhna Hai:

- 'Regulation of lac operon by repressor is referred to as' — know negative regulation
- lac operon gene-product mapping (Paper 2 match question with 'i, z, a, y' genes and products)
- induction by lactose/lactose analogues and role of lactose in switching on lac operon (Paper 4: 'addition of lactose regulates switching on')
- difference between positive and negative regulation terminology as tested in MCQs

Kaise Poochha Jaata Hai:

MCQ asking 'Regulation of lac operon by repressor is referred to as which type of regulation?' and matching/short-answer items linking genes to products; short descriptive 'Explain how addition of lactose regulates switching on of the lac operon'.

Repeat Pattern:

Consistently appears as MCQ and also short explanatory question — very likely to recur in similar formats.

Key Terms:

- *lac operon*
- *repressor*
- *negative regulation*
- *induction by lactose*

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5. Plasmid: definition as extrachromosomal circular DNA

High Conf. High Priority

Appeared in: Paper 1 · Paper 2 · Paper 3 · Paper 4 · 4× total

Question types: MCQ: 3 | Short: 1 | Long: 2

Kya Padhna Hai:

- 'Plasmid' defined as 'Extrachromosomal circular DNA' (MCQ exact wording)
- Role of plasmids as cloning vectors (pBR322 referenced across papers)
- Mapping of cloning sites and selectable markers on plasmid (pBR322: ori, rop, amp gene asked in Paper 2 & 3)
- Consequences of cloning vector lacking selectable marker (Paper 2 long question asks effect)

Kaise Poochha Jaata Hai:

MCQ identification, diagram-labelling long questions (draw pBR322 and label 'ori', 'rop', 'amp' etc.), and short essays about role and impact of markers.

Repeat Pattern:

Plasmid concept and pBR322 mapping recur in MCQs and Section E long questions — reliable high-yield topic.

Key Terms:

- *plasmid*
- *pBR322*
- *ori*
- *rop*
- *selectable marker*

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6. Restriction enzymes: recognition site naming and sticky/blunt ends

High Conf. High Priority

Appeared in: Paper 2 · Paper 4 · Paper 5 · Paper 3 · 4× total

Question types: MCQ: 2 | Short: 4 | Long: 1

Kya Padhna Hai:

- Naming rule for restriction endonuclease (Paper 5: 'How is a restriction endonuclease named?') with example EcoRI
- Recognise sequences and enzymes (Paper 2: '5'– G!"A A T T C –3' ... Name the restriction enzyme')
- Understand arrows indicating cleavage and resulting fragments (paper figures asked to 'What are the arrows indicating?')
- Distinguish sticky vs blunt ends and concept of insertional inactivation vs selectable markers (Paper 2 & 3 long Qs)

Kaise Poochha Jaata Hai:

Short numerical/identification: name enzyme that recognises sequence, what arrows indicate, result obtained; MCQ options with enzyme names; long-answer about cloning vector and insertional inactivation.

Repeat Pattern:

Recognition sequences + enzyme naming + cleavage consequence appear across multiple papers consistently.

Key Terms:

restriction endonuclease

EcoRI

recognition site

sticky ends

7. Replication fork: definition and role in replication of long DNA

High Conf. High Priority

Appeared in: Paper 1 · Paper 2 · Paper 3 · Paper 5 · 4× total

Question types: Short: 4

Kya Padhna Hai:

- 'What is a replication fork ? How does it aid replication of long strands of DNA ?' exact wording used in papers
- Mechanism: leading and lagging strands, continuous and discontinuous replication (Okazaki fragments)
- Diagrammatic representation asked in Paper 2: explain continuous/discontinuous with schematic
- Reason why replication occurs at forks not entire length simultaneously (space/enzymatic reasons as asked)

Kaise Poochha Jaata Hai:

Short-answer 3-mark questions asking definition + explanation; sometimes schematic required to explain continuous vs discontinuous replication.

Repeat Pattern:

Replication-fork phrasing is repeated across multiple papers in identical short-answer format — very likely to repeat.

Key Terms:

- replication fork
- leading strand
- lagging strand
- Okazaki fragments

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8. DNA polymorphism: definition and two applications

High Conf. High Priority

Appeared in: Paper 1 · Paper 3 · Paper 2 · 3× total

Question types: Short: 3

Kya Padhna Hai:

- 'What is polymorphism in DNA ? Give any two applications of DNA polymorphism.' exact wording used
- Core definition: presence of two or more alleles at a locus with appreciable frequency
- Two applications frequently expected: forensic identification (DNA fingerprinting), paternity testing, population genetic studies or disease allele mapping
- Example phrasing used in papers and expect to cite named applications

Kaise Poochha Jaata Hai:

Short 3-mark direct question: define term then give two applications — straight bullets expected.

Repeat Pattern:

Direct question repeats in at least three papers in identical wording !' high reliability.

Key Terms:

- DNA polymorphism
- forensic
- paternity
- population genetics

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9. PCR / molecular detection at low pathogen concentration

High Conf. High Priority

Appeared in: Paper 1 · Paper 2 · Paper 3 · Paper 5 · 4× total

Question types: Short: 4 | Long: 1

Kya Padhna Hai:

- Method that can detect pathogen at very low concentrations even when symptoms not visible: PCR (exact phrase used)
- Which two diseases detected by this method (papers asked to name two diseases e.g., HIV, Hepatitis C—typical examples)
- Principle of PCR basic steps (denaturation, annealing, extension) as required in longer biotech questions
- Importance of molecular probes and gel electrophoresis in detection pipeline (often paired)

Kaise Poochha Jaata Hai:

Short 3-mark question: 'Mention a method that can detect a pathogen at very low concentrations... Which two diseases are detected by this method ?' Expect method name + two examples.

Repeat Pattern:

PCR/molecular detection appears repeatedly across papers as a named method with disease examples !' high predictability.

Key Terms:

- PCR
- molecular diagnosis
- disease detection
- low concentration

10. Hardy-Weinberg equilibrium: definition and factors (Founder effect, gene migration)

High Conf. High Priority

Appeared in: Paper 1 · Paper 3 · Paper 4 · Paper 2 · 4× total

Question types: Short: 1 | Long: 1 | Case Study: 4

Kya Padhna Hai:

- Exact definition asked: 'Define Hardy-Weinberg equilibrium.' (paper wording)
- Founder effect: 'What do you understand by Founder effect ?' with examples of small founder populations changing allele frequencies
- Gene migration: how migration affects equilibrium (paper asked 'Gene migration can affect Hardy-Weinberg')
- Know list of factors affecting equilibrium: gene migration, genetic drift, mutation, selection, non-random mating

Kaise Poochha Jaata Hai:

Case-based 4-mark passage: define H-W equilibrium (a), explain Founder effect (b), and either explain gene migration effect or how mutations result in evolution (c).

Repeat Pattern:

Hardy-Weinberg and associated factors appear as a dedicated case each year in at least three papers — very high recurrence in case format.

Key Terms:

- Hardy-Weinberg
- Founder effect
- gene migration
- genetic drift

11. Population logistic growth model: equation and K carrying capacity

High Conf. | High Priority

Appeared in: Paper 1 · Paper 2 · Paper 4 · Paper 3 · 4× total

Question types: Short: 4 | Long: 1

Kya Padhna Hai:

- 'Describe the population logistic growth model and provide its equation.' (paper phrase)
- Know Verhulst-Pearl logistic equation: dN/dt = rN((K-N)/K) and definition of K as carrying capacity ('What does K represent?')
- Be able to compute N(t+1) with natality/mortality/immigration/emigration numbers (Paper 1 & 3 style numeric question)
- Understand difference between exponential (rN) and logistic growth (rN((K-N)/K)) for graphs

Kaise Poochha Jaata Hai:

Short 2–5 mark tasks: write equation, define K, and numeric calculation 'If N=500 then what is N(t+1) given natality etc.'

Repeat Pattern:

Logistic model equation and K asked across multiple papers including calculation tasks — highly recurring.

Key Terms:

- *logistic growth*
- *dN/dt = rN((K-N)/K)*
- *carrying capacity*
- *K*

12. Uterus wall structure: layers and functions of any two layers

High Conf. | High Priority

Appeared in: Paper 1 · Paper 3 · Paper 4 · 3× total

Question types: Short: 3

Kya Padhna Hai:

- 'Describe the structure of the uterus wall and functions performed by any of its two layers.' (paper exact wording)
- Know layers: perimetrium (outer), myometrium (middle muscular), endometrium (inner) and specific functions like implantation site, shedding during menstruation, contractions during labour
- Role of endometrium sub-layers (functional vs basal) if asked
- Example phrasing expects naming two layers and giving their functions concisely

Kaise Poochha Jaata Hai:

3-mark short question: draw or describe layers and list functions of two; expect one-sentence function per layer.

Repeat Pattern:

Uterus wall + functions asked multiple times across papers in short-answer format !’ reliable.

Key Terms:

- *endometrium*
- *myometrium*
- *perimetrium*
- *implantation*

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13. Ploidy of primary and secondary spermatocytes

High Conf. High Priority

Appeared in: Paper 1 · Paper 3 · Paper 5 · 3× total

Question types: Short: 3

Kya Padhna Hai:

- 'What is the ploidy of primary spermatocytes and secondary spermatocytes ?' exact short-question phrasing
- Know primary spermatocyte is diploid (2n) and secondary spermatocyte is haploid (n)
- Understand stage mapping: spermatogonium -> primary spermatocyte -> meiosis I -> secondary spermatocyte -> meiosis II -> spermatids
- Be ready to state ploidy succinctly as required for 2-mark answers

Kaise Poochha Jaata Hai:

Short 2-mark definition-style question requiring precise ploidy terms (2n vs n) and possibly brief stage mapping.

Repeat Pattern:

Short ploidy questions recur often in Section B — high chance.

Key Terms:

- *primary spermatocyte*
- *secondary spermatocyte*
- *2n*
- *n*

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14. Blood groups: non-Mendelian patterns (multiple allelism and co-dominance)

High Conf. High Priority

Appeared in: Paper 1 · Paper 2 · Paper 4 · Paper 5 · 4× total

Question types: Short: 4 | Long: 1

Kya Padhna Hai:

- 'Explain briefly the two non-Mendelian patterns of inheritance that are observed in blood groups of humans.' exact phrasing in Paper 1
- Concepts: multiple allelism (IA, IB, i) and co-dominance (IA and IB expressed together as AB)
- Be prepared to explain ABO system with genotype-phenotype mapping and example Punnett outcomes
- If asked, tie into clinical relevance: transfusion compatibility / antibody presence

Kaise Poochha Jaata Hai:

Short 2–3 mark direct question: name two patterns and briefly explain with examples (ABO).

Repeat Pattern:

Blood group non-Mendelian patterns are repeatedly tested in Section B/C — staple topic.

Key Terms:

- ABO system
- multiple allelism
- co-dominance
- IA IB i

15. Methods to introduce alien DNA into animal and plant cells (electroporation, biolistic/gene gun)

High Conf. | High Priority

Appeared in: Paper 1 · Paper 2 · Paper 4 · Paper 5 · 4× total

Question types: Short: 4 | Long: 2

Kya Padhna Hai:

- Paper phrasing: 'Name and explain one method which is used to introduce alien DNA into an animal cell and plant cell respectively.'
- Methods to know: microinjection/electroporation for animal cells; biolistic (gene gun) and Agrobacterium-mediated for plant cells (Paper 2 specifically names 'biolistic gun')
- Understand mechanism briefly: how gene gun propels DNA-coated particles, how electroporation increases membrane permeability
- Be ready to mention one example per cell-type as short explanation

Kaise Poochha Jaata Hai:

2-mark short 'name and explain one method' format; longer biotech questions expect mechanism steps.

Repeat Pattern:

This exact wording method-question recurs across many papers — high expectation to prepare both animal and plant techniques.

Key Terms:

- electroporation
- biolistic gun
- Agrobacterium
- microinjection

16. Pedigree chart interpretation: inheritance pattern identification (X-linked, autosomal)

High Conf. High Priority

Appeared in: Paper 2 · Paper 4 · Paper 5 · Paper 3 · 4× total

Question types: MCQ: 3 | Short: 2 | Long: 1

Kya Padhna Hai:

- 'Study the pedigree chart of a family ... identify the nature of the trait depicted'—be ready to classify: dominant/recessive, autosomal/X-linked
- Recognise common pedigree features: affected males only vs both sexes, carrier females, father-to-son transmission absent/present
- Map symbols and inheritance rules quickly to choose between options
- Practice with sample pedigrees as MCQs used diagrams

Kaise Poochha Jaata Hai:

MCQ with pedigree diagram and options (Dominant X-linked / Recessive X-linked / Autosomal dominant / Autosomal recessive).

Repeat Pattern:

Pedigree interpretation is a staple MCQ across papers.

Key Terms:

- *pedigree*
- *X-linked*
- *autosomal dominant*
- *carrier*

17. Vaccination long-term immunity: immunological memory and anamnestic response

High Conf. High Priority

Appeared in: Paper 1 · Paper 3 · Paper 2 · Paper 5 · 4× total

Question types: Short: 2 | Long: 1 | Case Study: 4

Kya Padhna Hai:

- Exact question wording: 'What name is given to such a type of immune response ? Also name the property which is responsible for it.' (papers)
- Terms: 'memory response' / 'immunological memory' and 'anamnestic response' definitions
- Types of immune memory: humoral (memory B-cells) and cell-mediated (memory T-cells) often asked
- Special cells helping long-term immunity: memory B cells and plasma cells; mechanism of secondary response (faster, higher antibody titer)

Kaise Poochha Jaata Hai:

Case-based short parts: define type of response, name property responsible, describe types, and explain role of special blood cells or define anamnestic response.

Repeat Pattern:

Repeated case passage asking identical subparts across multiple papers !’ very high predictability.

Key Terms:

- *memory response*
- *anamnestic response*
- *memory B cells*
- *secondary immune response*

18. Angiosperm seed advantages and polyembryony (define and example)

High Conf. | High Priority

Appeared in: Paper 1 · Paper 3 · Paper 2 · 3× total

Question types: Short: 1 | Long: 3

Kya Padhna Hai:

- 'Describe how the presence of seed is advantageous to angiosperms. Mention any three points.' exact Paper phrasing
- Specific advantages to list: protection of embryo, dispersal mechanisms, nourishment (endosperm), dormancy capability, wider distribution
- 'What is Polyembryony ? Give an example.' — define polyembryony and give example like citrus or certain plants cited in papers
- Differentiate perisperm vs pericarp when asked (Paper 1 had differentiation item)

Kaise Poochha Jaata Hai:

Long 5-mark multipart: list three advantages, differentiate perisperm vs pericarp, and define polyembryony with example.

Repeat Pattern:

Angiosperm seed advantage and polyembryony are standard Section E topics repeated across papers.

Key Terms:

- *seed advantage*
- *polyembryony*
- *perisperm*
- *pericarp*

19. AIDS: cells attacked (CD4+ T cells), diagnostic tests (ELISA), and WHO prevention steps

High Conf. | High Priority

Appeared in: Paper 5 · Paper 4 · Paper 2 · 3× total

Question types: MCQ: 1 | Short: 3 | Case Study: 4

Kya Padhna Hai:

- 'Name the group of cells the HIV attacks after gaining entry into the human body' — CD4+ T helper lymphocytes (exact phrasing used)
- Diagnostic test expanded form: ELISA (Enzyme-Linked Immunosorbent Assay) as asked in Paper 5
- Possible treatments mentioned historically: antiretroviral therapy (ART) — list as 'possible treatment available'
- WHO preventive steps: safe sex, needle exchange, blood screening, mother-to-child transmission prevention;

papers asked for 'any two steps suggested by WHO for preventing the spread'

Kaise Poochha Jaata Hai:

Case-based question with short subparts: name attacked cell, expand diagnostic test acronym, list treatments, mention WHO prevention steps or explain why AIDS patients die of opportunistic infections.

Repeat Pattern:

AIDS case appears in multiple papers with very similar subparts; highly probable in exam sets.

Key Terms:

- HIV
- CD4+ T cells
- ELISA
- ART

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20. Spermatogenesis: stages, hormones (FSH, LH) and cell-stage mapping

High Conf. High Priority

Appeared in: Paper 5 · Paper 4 · Paper 1 · Paper 3 · 4× total

Question types: MCQ: 1 | Short: 3 | Long: 2 | Case Study: 4

Kya Padhna Hai:

- 'Describe the events of spermatogenesis with the help of a schematic diagram.' (Paper 2/5 phrasing)
- Site of action of FSH (Sertoli cells) and its actions (spermiation support, nourishment) and role of LH (Leydig cells !' testosterone) as asked
- Name cells/products undergoing mitosis and meiosis (spermatogonia, primary spermatocyte, secondary spermatocyte, spermatids)
- Accessory ducts sperms travel through from seminiferous tubules to epididymis (rete testis, efferent ductules) as Paper 5 asked

Kaise Poochha Jaata Hai:

Long diagram+explain questions and short 2-mark questions about hormone sites and cell-stage mapping; also OR choices on role of LH/FSH.

Repeat Pattern:

Spermatogenesis + hormonal regulation is heavily repeated across papers in Section E and B !' very high-value topic.

Key Terms:

- FSH
- LH
- Sertoli cells
- Leydig cells

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21. Events in mosquito after sucking infected blood

High Conf. High Priority

Appeared in: Paper 2 · 1× total

Question types: MCQ: 0 | Short: 1 | Long: 0 | Case Study: 1

Kya Padhna Hai:

- Memorise the sequence of Plasmodium development stages inside the female Anopheles after it sucks infected blood as asked in the paper: uptake of gametocytes, gametogenesis in mosquito gut, fusion to form zygote, ookinete formation and penetration of gut wall, oocyst formation on midgut wall, sporozoite formation and migration to salivary glands.
- Be ready to state this as a short ordered list in 2–3 lines because the paper asks: 'Explain the events which occur within a female Anopheles mosquito after it has sucked blood from a malaria patient.'
- Know the forms entering human (sporozoites) vs those entering mosquito (gametocytes) to answer linked subparts.

Kaise Poochha Jaata Hai:

Short 2-mark question in a case about malaria vectors: expect to list sequential developmental events inside the mosquito (from gametocytes to sporozoites).

Repeat Pattern:

Case-based passage on malaria in Paper 2 demanded this exact sequence — practise writing steps in order concisely.

Key Terms:

- *gametocytes*
- *gametogenesis*
- *zygote*
- *ookinete*
- *oocyst*
- *sporozoite*
- *salivary glands*

22. Species-Area relationship and logistic growth equation

High Conf. High Priority

Appeared in: Paper 4 · Paper 2 · 2× total

Question types: MCQ: 0 | Short: 1 | Long: 2 | Case Study: 0

Kya Padhna Hai:

- Learn Alexander von Humboldt's Species-Area relationship formula and form: S = C A^Z (know meaning of S = species number, A = area, C and Z constants) and be able to sketch the curve (paper asks to 'Draw the graph showing Species-Area relationship for S = C A^Z').
- Review the Verhulst-Pearl logistic growth equation and its form dN/dt = rN((K-N)/K), know what each symbol means and be able to explain the S-shaped logistic curve and role of carrying capacity K; the paper explicitly asks for the equation and graphical representation.
- Practice interpreting graphs: given curves identify logistic vs exponential, and compute simple numerical population change when natality/mortality/immigration/emigration values are provided (papers include such numeric tasks).

- Be prepared to state ecological implications: how increasing area increases species number (species-area), and how carrying capacity limits growth (logistic).

Kaise Poochha Jaata Hai:

Section E long question may ask to 'Describe Species-Area relationship' including graph; separate short/case questions ask to 'Describe logistic population growth curve' and 'Write the equation of Verhulst-Pearl logistic growth curve'. Numeric short problems may follow.

Repeat Pattern:

Paper 4 explicitly contains both species-area and logistic growth items — expect textbook formulae, sketching ability, and interpretation.

Key Terms:

- Species-Area relationship
- S = C A^Z
- logistic growth
- dN/dt = rN((K-N)/K)
- carrying capacity K

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23. IUD types: Multiload 375 and copper IUDs (named brand mention)

Medium Conf. | Medium Priority

Appeared in: Paper 1 · Paper 5 · 2× total

Question types: MCQ: 2 | Short: 1

Kya Padhna Hai:

- Recognise 'Multiload 375' as an IUD used by females (MCQ exact phrasing)
- Know examples of copper-releasing intra-uterine devices (Paper 5 asked to 'Name any two copper releasing intra-uterine devices')
- Two reasons why copper IUDs are effective as contraceptives (as requested in Paper 5)
- Distinguish IUD from diaphragm and vault (MCQ distractors used)

Kaise Poochha Jaata Hai:

Short MCQ 'IUD used by females as contraceptive device is :' and direct short answer 'Name any two copper releasing intra-uterine devices. State two reasons'.

Repeat Pattern:

Named device Multiload 375 appears in MCQs in multiple papers; specifics (names + reasons) asked at least once !' moderate predictability.

Key Terms:

- Multiload 375
- copper IUD
- contraceptive devices
- effectiveness reasons

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24. RNA as catalytic molecule (ribozymes) / first genetic material evidence

Medium Conf. Medium Priority

Appeared in: Paper 1 · Paper 2 · 2× total

Question types: MCQ: 2

Kya Padhna Hai:

- 'RNA can act as catalysts' used as correct evidence for 'RNA as first genetic material' in MCQ
- Know ribozyme concept and examples (papers used 'RNA can act as catalysts')
- Distinguish RNA stability and double/single stranded statements (MCQ distractors used)
- Understand why RNA is considered in 'RNA world' hypothesis at MCQ level

Kaise Poochha Jaata Hai:

MCQ: pick correct statement among options about RNA being first genetic material using 'RNA can act as catalysts' phrasing.

Repeat Pattern:

Appears as MCQ phrasing across at least two papers; expect similar one-liners.

Key Terms:

ribozyme

RNA world

catalysis

first genetic material

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25. Baculoviruses as biological control: three reasons they are preferred

Medium Conf. Medium Priority

Appeared in: Paper 1 · Paper 3 · 2× total

Question types: Short: 2

Kya Padhna Hai:

- Exact question: 'Though baculoviruses are pathogens, they are used as biological control agents. Give three reasons why they are preferred.'
- Typical concrete reasons expected: host-specificity to insect pests; safety to vertebrates and plants; ability to be formulated and sprayed; persistence in environment — use wording from paper
- Provide three named reasons with brief justification (each short bullet) as papers expect
- Know example application (e.g., control of lepidopteran pests) if asked

Kaise Poochha Jaata Hai:

Short 3-mark 'give three reasons' format — list three concise points, each clearly tied to baculovirus properties.

Repeat Pattern:

Appeared twice across papers in same short-answer format — moderately repeatable.

Key Terms:

baculovirus

biological control

host-specificity

safety

26. Decomposition to humus: stepwise process (fragmentation, leaching, catabolism, mineralisation, humification)

Medium Conf. Medium Priority

Appeared in: Paper 2 · Paper 1 · 2× total

Question types: Short: 2

Kya Padhna Hai:

- Exact step list used in Paper 2: 'Fragmentation, Leaching, Catabolism, Mineralisation, Humification' and order of occurrence
- Be ready to 'Explain the fourth step in the process' (mineralisation in that ordering) with examples
- Definition of humus and its properties (resistant to microbial action, colloidal) as contrasted in MCQ
- Biological agents and factors favouring each step (temperature, detritus nature) if short justification is asked

Kaise Poochha Jaata Hai:

Short 2–3 mark questions: order the steps and explain a specified step (paper asked to 'explain the fourth step').

Repeat Pattern:

Decomposition step-order and humus properties appear in MCQ/short-answer across papers occasionally !' moderate chance to repeat.

Key Terms:

fragmentation

leaching

catabolism

mineralisation

humification

27. Trophoblast function in human embryo

Medium Conf. Medium Priority

Appeared in: Paper 1 · Paper 3 · 2× total

Question types: Short: 2

Kya Padhna Hai:

- Exact paper phrasing: 'Mention the function of trophoblast in the human embryo.'
- Know trophoblast roles: formation of placenta, invasion of uterine wall, secretion of hCG (supporting corpus luteum)
- If asked, link trophoblast to formation of chorionic villi and maternal-fetal exchange
- Expect 1–2 brief points; paper asks succinctly in 2-mark format

Kaise Poochha Jaata Hai:

2-mark short direct question: name function(s) — write 2 concise roles (placenta formation, hCG secretion).

Repeat Pattern:

Appears in Section B as an OR alternative across multiple papers — moderate recurrence.

Key Terms:

- trophoblast
- placenta formation
- hCG
- chorionic villi

28. Polygenic inheritance deviation from Mendelian patterns

Medium Conf. | Medium Priority

Appeared in: Paper 1 · Paper 3 · 2× total

Question types: Short: 2

Kya Padhna Hai:

- Paper wording: 'How does polygenic inheritance show deviation from Mendelian inheritance ?'
- Points expected: multiple genes contribute to single trait, continuous variation, quantitative traits like skin colour/height
- Contrast with Mendel's discrete trait segregation and fixed ratios
- Give one concrete human example (height, skin colour) as often expected

Kaise Poochha Jaata Hai:

2-mark short explanation: state how polygenic differs and provide brief example.

Repeat Pattern:

Non-Mendelian inheritance concepts (polygenic) appear as OR options in Section B regularly.

Key Terms:

- polygenic
- continuous variation
- quantitative trait
- Mendelian deviation

29. Tears as immune barrier: chemical barrier example and analogues

Medium Conf. | Medium Priority

Appeared in: Paper 1 · Paper 3 · 2× total

Question types: Short: 2

Kya Padhna Hai:

- Exact question: 'What function is performed by tears shed by our eyes ? Which type of barrier is this in the process of immunity ? Give another similar example.'
- Know tears contain lysozyme—chemical barrier; function: wash away microbes and contain antimicrobial enzymes
- Give another similar example asked: saliva or mucus in respiratory tract as chemical barrier
- Prepare concise phrasing: function + barrier type + one analogous example

Kaise Poochha Jaata Hai:
2-mark question: short sentence for function, name barrier type (chemical), and name one similar example.

Repeat Pattern:
Barrier-type examples (tears) appear a few times across papers in identical short-answer style.

Key Terms:

- *tears*
- *lysozyme*
- *chemical barrier*
- *saliva*

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30. Adaptive radiation / divergent evolution identification and examples

Medium Conf. | Medium Priority

Appeared in: Paper 1 · Paper 5 · 2× total

Question types: MCQ: 2 | Short: 1 | Long: 0 | Case Study: 0

Kya Padhna Hai:
- Recognise the definition used in MCQ stems: 'Species which have diverged after origin from common ancestor giving rise to new species, adapted to new habitats and ways of life is called :' (adaptive radiation/ divergent evolution). Read and memorise the exact phrase used in the paper.
- Learn 3 concrete examples of adaptive radiation (e.g., Darwin's finches, Hawaiian honeycreepers, African Rift Lake cichlids) so you can cite at least one when asked for an example.
- Understand the distinction between adaptive radiation/divergent evolution and convergent evolution (adaptive radiation = diversification from common ancestor into many forms adapted to different niches; convergent = unrelated lineages evolve similar traits).
- Be prepared to recognise short-definition stems and to give one-line explanations and one example if asked in short-answer format.

Kaise Poochha Jaata Hai:
Usually a one-line MCQ asking to pick the correct evolutionary term that fits the given description. Occasionally the concept may be part of a longer evolution question.

Repeat Pattern:
Exact definitional MCQ stems recur; knowing one clear example plus the textbook distinction is sufficient to answer both MCQ and a brief short-answer.

Key Terms:

- *adaptive radiation*
- *divergent evolution*
- *convergent evolution*
- *example: Darwin's finches*

31. Cloning vector pBR322: diagram labelling and role of 'rop' gene

Medium Conf. | Medium Priority

Appeared in: Paper 2 · Paper 3 · 2× total

Question types: Long: 2

Kya Padhna Hai:

- 'Draw a schematic diagram of the cloning vector pBR 322 and label (1) Bam HI site (2) gene for ampicillin resistance (3) ori (4) rop gene.' exact phrasing
- Role of 'rop' gene: regulation of plasmid copy number (papers asked explicitly)
- Consequences when cloning vector lacks selectable marker (how cloning process affected) and insertional inactivation preference
- Be ready to discuss selectable marker vs insertional inactivation pros/cons

Kaise Poochha Jaata Hai:

Long 5-mark diagram + short descriptive parts: label diagram, state role, and explain marker consequences/ insertional inactivation.

Repeat Pattern:

pBR322 diagram/rop role repeated in Section E across papers — expect similar.

Key Terms:

pBR322

BamHI

ampicillin resistance

rop gene

32. Predator vs prey: distinction and ecological roles of predators

Medium Conf. | Medium Priority

Appeared in: Paper 1 · Paper 3 · Paper 5 · 3× total

Question types: Short: 3

Kya Padhna Hai:

- 'Distinguish between predator and prey. Mention any two significant roles that predators play in nature.' exact wording used
- Concrete distinguishing points: trophic level roles, mode of obtaining food (active hunting vs being consumed)
- Two roles expected: population control of prey, maintaining species diversity, removing weak/old individuals—give specific examples like lion-herbivore dynamics
- Prepare two concise role explanations with named examples

Kaise Poochha Jaata Hai:

3-mark short question: define both terms briefly and list two roles of predators with one-sentence justification.

Repeat Pattern:

Appears in Section C across papers; moderate repeatability.

Key Terms:

- *predator*
- *prey*
- *population control*
- *biodiversity*

33. In situ vs ex situ conservation: differences and examples

Medium Conf. Medium Priority

Appeared in: Paper 1 · Paper 4 · 2× total

Question types: Short: 2 | Long: 1

Kya Padhna Hai:

- 'Differentiate between in situ conservation and ex situ conservation.' (paper phrasing)
- Concrete points: in situ = in natural habitats (national parks, biosphere reserves) vs ex situ = outside natural habitat (botanical gardens, seed banks, zoos)
- Provide examples: Kaziranga (in situ), seed bank/CBD/zoos for ex situ
- Explain advantages/limitations as asked in 5-mark form if required

Kaise Poochha Jaata Hai:

Direct 5-mark or 2-mark comparison: list differences and give specific examples.

Repeat Pattern:

Conservation compare questions appear occasionally in Section E and C; prepare both definitions and examples.

Key Terms:

- *in situ*
- *ex situ*
- *seed bank*
- *national park*

Related Topic Pairs

Yeh topics exam mein aksar ek saath poochhe jaate hain — inhe milake padho.

1. Cloning vector pBR322 & Restriction enzymes: often combined in Section E diagram + discussion where enzyme sites (BamHI, EcoRI) are mapped on pBR322 and role of insertional inactivation/selectable markers is discussed.

2. Replication fork & Okazaki fragments (Replication fork question usually paired with continuous/discontinuous replication schematic) — students should prepare both together.

3. PCR / Gel electrophoresis: molecular detection question (PCR) is frequently paired with gel electrophoresis as the downstream separation/visualisation step in long-answer biotech questions.

Apni Strategy Chuno

Pick your goal — here's exactly which topics to study based on past paper patterns.

🔴 Bas Pass Hona Hai

Just want to pass — 22 of 33 topics · ~77% marks coverage

1. Stop codon identification (UGA as stop codon)
2. Chargaff's rule / base proportion in double-stranded DNA
3. lac operon regulation by repressor (negative regulation)
4. Plasmid: definition as extrachromosomal circular DNA
5. Restriction enzymes: recognition site naming and sticky/blunt ends
6. Replication fork: definition and role in replication of long DNA
7. PCR / molecular detection at low pathogen concentration
8. Hardy-Weinberg equilibrium: definition and factors (Founder effect, gene migration)
9. Population logistic growth model: equation and K carrying capacity
10. Blood groups: non-Mendelian patterns (multiple allelism and co-dominance)
11. Methods to introduce alien DNA into animal and plant cells (electroporation, biolistic/gene gun)
12. Pedigree chart interpretation: inheritance pattern identification (X-linked, autosomal)
13. Vaccination long-term immunity: immunological memory and anamnestic response
14. Spermatogenesis: stages, hormones (FSH, LH) and cell-stage mapping
15. Microspore formation: meiosis of pollen mother cell
16. DNA polymorphism: definition and two applications
17. Uterus wall structure: layers and functions of any two layers
18. Ploidy of primary and secondary spermatocytes
19. Angiosperm seed advantages and polyembryony (define and example)
20. AIDS: cells attacked (CD4+ T cells), diagnostic tests (ELISA), and WHO prevention steps
21. Species-Area relationship and logistic growth equation
22. Events in mosquito after sucking infected blood

Average Score Chahiye

Decent score — 33 of 33 topics · ~100% marks coverage

1. Stop codon identification (UGA as stop codon)
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22. Events in mosquito after sucking infected blood
23. Predator vs prey: distinction and ecological roles of predators
24. IUD types: Multiload 375 and copper IUDs (named brand mention)
25. RNA as catalytic molecule (ribozymes) / first genetic material evidence
26. Baculoviruses as biological control: three reasons they are preferred
27. Decomposition to humus: stepwise process (fragmentation, leaching, catabolism, mineralisation, humification)
28. Trophoblast function in human embryo
29. Polygenic inheritance deviation from Mendelian patterns
30. Tears as immune barrier: chemical barrier example and analogues
31. Adaptive radiation / divergent evolution identification and examples
32. Cloning vector pBR322: diagram labelling and role of 'rop' gene
33. In situ vs ex situ conservation: differences and examples

🎯 Top Karna Hai

Full coverage — all 33 topics · 100% marks coverage

1. Stop codon identification (UGA as stop codon)
2. Chargaff's rule / base proportion in double-stranded DNA
3. lac operon regulation by repressor (negative regulation)
4. Plasmid: definition as extrachromosomal circular DNA
5. Restriction enzymes: recognition site naming and sticky/blunt ends
6. Replication fork: definition and role in replication of long DNA
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33. In situ vs ex situ conservation: differences and examples

Overall Exam Strategy

Bas Pass Hona Hai: agar tum sirf kuch granular cheezen padhoge jo 40-50% questions cover kar denge, toh yeh topics zaroor cover karo: "Chargaff's rule / base composition", "Stop codon identification (UGA)", "lac operon regulation by repressor", "Plasmid & pBR322 (ori, rop, amp) and restriction enzymes (EcoRI etc.)", "Replication fork — leading/lagging and Okazaki fragments", "PCR and gel electrophoresis (ethidium bromide + principle)", "Hardy-Weinberg equilibrium + Founder effect + simple allele frequency calculation", "Population logistic model and K calculation". Kyun: yeh set baar-baar MCQ/short/case formats mein repeat hua hai — together in papers yeh 8–10 granular topics let you answer multiple MCQs, several 2–3 mark short answers, one case on H-W, plus a numerical on population growth and a common biotech long question. Agar tum in topics par crisp definitions, one-line mechanisms/steps, one worked numeric example, aur pBR322/gel diagrams practice kar loge, toh tum conservative estimate se 40–50% questions confidently attempt kar loge.

Exam tip: MCQ aur 2-mark short answers ko pehle khao — bahut se papers repeat karte hain identical wording; diagrams (pBR322, replication fork, spermatogenesis) practice karke time bachaoge. Prioritise memory of exact phrases like 'Microspores are formed by', 'Which of the following options is a Stop codon ?', 'Define Hardy-Weinberg equilibrium' — papers often repeat stems verbatim.